

## CABO CARVOEIRO, Portugal

WMO: 085310

Lat: 39.3614N

Lon: 9.4069W

Elev: 105

StdP: 14.64

Time Zone: 0.00 (EUW)

Period: 90-18

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 43.9       | 46.1 | 24.4                          | 18.6 | 53.1 | 30.9 | 25.2 | 49.2 | 33.2                  | 53.5 | 29.5 | 50.7 | 12.3                  | 90   | 0.489 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 5.3                    | 73.9            | 65.5 | 71.7 | 65.9 | 70.2 | 65.5 | 68.5                | 71.2 | 67.5 | 70.0 | 66.7 | 69.1 | 8.0                  | 230  |

| Dehumidification DP/MCDB and HR |       |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 67.5                            | 101.5 | 69.9 | 66.6 | 98.2 | 69.1 | 65.7 | 95.2 | 68.4 | 32.8          | 71.4 | 32.0 | 70.2 | 31.3 | 69.4 | 74.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |    |    |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|----|----|
|                   |      |      | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |    |    |
| (4) 27.6          | 24.6 | 22.1 | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 60.4   | 54.4  | 54.1  | 56.5  | 58.3  | 61.3  | 64.0  | 65.9  | 66.2  | 65.7  | 63.6  | 59.0  | 55.3  |
|                                                                        | DBStd       | 5.39   | 3.36  | 3.34  | 3.18  | 3.41  | 3.11  | 2.30  | 2.01  | 2.06  | 2.37  | 3.24  | 3.68  | 3.52  |
|                                                                        | HDD50       | 21     | 8     | 8     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 5     |
|                                                                        | HDD65       | 1853   | 330   | 304   | 265   | 205   | 121   | 46    | 11    | 6     | 19    | 63    | 182   | 301   |
|                                                                        | CDD50       | 3815   | 143   | 124   | 201   | 249   | 352   | 419   | 493   | 503   | 470   | 423   | 269   | 169   |
|                                                                        | CDD65       | 172    | 0     | 0     | 1     | 3     | 8     | 15    | 39    | 45    | 39    | 21    | 1     | 0     |
|                                                                        | CDH74       | 84     | 0     | 0     | 1     | 3     | 15    | 2     | 12    | 21    | 11    | 19    | 0     | 0     |
|                                                                        | CDH80       | 7      | 0     | 0     | 0     | 0     | 2     | 0     | 0     | 3     | 1     | 1     | 0     | 0     |
| (14) Wind                                                              | WSAvg       | 11.0   | 11.7  | 12.5  | 12.9  | 11.5  | 10.6  | 10.2  | 10.3  | 10.3  | 9.5   | 9.3   | 11.6  | 12.0  |
| (15) Precipitation                                                     | PrecAvg     | 23.6   | 3.3   | 3.0   | 1.8   | 2.0   | 1.7   | 0.7   | 0.2   | 0.2   | 0.9   | 2.8   | 3.6   | 3.4   |
|                                                                        | PrecMax     | 35.3   | 10.4  | 9.6   | 5.4   | 5.4   | 4.6   | 2.6   | 1.2   | 2.0   | 3.8   | 7.6   | 11.3  | 10.3  |
|                                                                        | PrecMin     | 14.4   | 0.0   | 0.2   | 0.0   | 0.3   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.1   | 0.0   | 0.2   |
|                                                                        | PrecStd     | 5.6    | 2.1   | 2.0   | 1.4   | 1.3   | 1.3   | 0.7   | 0.3   | 0.4   | 1.0   | 2.2   | 2.4   | 2.6   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 61.9  | 64.1  | 71.0  | 73.3  | 75.0  | 73.3  | 75.5  | 75.9  | 75.1  | 76.6  | 68.5  | 63.8  |
|                                                                        |             | MCWB   | 57.8  | 55.9  | 57.6  | 60.4  | 63.9  | 66.8  | 68.6  | 68.0  | 67.4  | 62.4  | 61.7  | 59.4  |
|                                                                        | 2%          | DB     | 60.4  | 61.1  | 65.2  | 68.2  | 69.8  | 70.1  | 71.7  | 72.3  | 72.0  | 71.5  | 66.3  | 62.1  |
|                                                                        |             | MCWB   | 56.4  | 56.0  | 57.0  | 61.1  | 63.0  | 65.2  | 67.1  | 67.7  | 66.9  | 63.1  | 62.1  | 59.0  |
|                                                                        | 5%          | DB     | 59.6  | 59.6  | 62.3  | 64.8  | 67.1  | 68.4  | 70.0  | 70.5  | 70.3  | 69.2  | 64.9  | 61.1  |
|                                                                        |             | MCWB   | 55.7  | 55.8  | 56.3  | 60.1  | 62.0  | 64.1  | 66.2  | 66.6  | 66.4  | 63.9  | 60.9  | 58.1  |
|                                                                        | 10%         | DB     | 58.8  | 58.5  | 60.4  | 62.7  | 65.3  | 67.2  | 68.7  | 69.3  | 69.2  | 67.8  | 63.9  | 60.1  |
|                                                                        |             | MCWB   | 55.0  | 55.2  | 55.8  | 59.1  | 61.2  | 63.2  | 65.2  | 65.8  | 65.8  | 63.8  | 60.0  | 56.8  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 60.0  | 59.1  | 61.1  | 64.4  | 66.3  | 67.8  | 69.7  | 70.6  | 69.9  | 67.8  | 64.2  | 61.4  |
|                                                                        |             | MCDB   | 60.9  | 60.9  | 65.1  | 69.3  | 70.8  | 71.9  | 73.1  | 74.0  | 73.5  | 69.5  | 66.1  | 62.6  |
|                                                                        | 2%          | WB     | 58.7  | 57.9  | 59.2  | 61.7  | 64.2  | 66.2  | 68.1  | 68.4  | 68.1  | 66.5  | 62.7  | 60.2  |
|                                                                        |             | MCDB   | 60.0  | 59.9  | 61.8  | 65.7  | 67.4  | 69.3  | 70.5  | 71.0  | 70.5  | 69.1  | 64.7  | 61.5  |
|                                                                        | 5%          | WB     | 57.5  | 56.8  | 57.9  | 60.4  | 62.8  | 64.7  | 67.0  | 67.3  | 67.2  | 65.5  | 61.6  | 59.1  |
|                                                                        |             | MCDB   | 59.2  | 58.9  | 60.5  | 64.2  | 66.1  | 67.5  | 69.4  | 69.6  | 69.5  | 68.2  | 63.7  | 60.7  |
|                                                                        | 10%         | WB     | 56.4  | 55.7  | 56.7  | 59.2  | 61.7  | 63.8  | 65.7  | 66.4  | 66.3  | 64.4  | 60.5  | 58.0  |
|                                                                        |             | MCDB   | 58.4  | 58.1  | 59.5  | 62.5  | 64.9  | 66.5  | 68.3  | 68.6  | 68.8  | 67.0  | 62.9  | 60.0  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 5.5   | 6.1   | 6.1   | 5.5   | 5.2   | 4.7   | 4.8   | 5.3   | 5.3   | 6.4   | 5.7   | 6.3   |
|                                                                        |             | MCDBR  | 5.7   | 8.0   | 11.0  | 10.0  | 9.4   | 7.0   | 7.0   | 7.9   | 7.5   | 9.6   | 6.3   | 6.4   |
|                                                                        |             | MCWBR  | 5.3   | 6.1   | 7.6   | 6.0   | 5.2   | 4.2   | 4.4   | 4.4   | 4.8   | 6.5   | 6.3   | 6.1   |
|                                                                        | 5% WB       | MCDBR  | 4.3   | 6.2   | 7.7   | 8.6   | 7.9   | 6.5   | 6.5   | 6.7   | 7.3   | 8.1   | 5.3   | 5.0   |
|                                                                        |             | MCWBR  | 4.9   | 5.6   | 6.8   | 5.6   | 5.1   | 4.3   | 4.7   | 4.6   | 5.0   | 5.8   | 5.3   | 5.9   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.334 | 0.345 | 0.376 | 0.382 | 0.389 | 0.394 | 0.381 | 0.392 | 0.391 | 0.369 | 0.343 | 0.333 |
|                                                                        | taud        |        | 2.506 | 2.472 | 2.393 | 2.382 | 2.394 | 2.399 | 2.430 | 2.408 | 2.423 | 2.491 | 2.517 | 2.505 |
|                                                                        | Ebn at Noon |        | 265   | 278   | 279   | 284   | 283   | 280   | 283   | 277   | 271   | 266   | 259   | 256   |
|                                                                        | Edh at Noon |        | 25    | 29    | 35    | 37    | 37    | 37    | 36    | 36    | 33    | 28    | 24    | 23    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 674   | 992   | 1357  | 1770  | 2102  | 2231  | 2250  | 2070  | 1619  | 1121  | 755   | 609   |
|                                                                        | RadStd      |        | 85    | 114   | 143   | 147   | 148   | 108   | 106   | 87    | 100   | 83    | 92    | 66    |

## Historical Trends

|                              | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (17 neighbors) | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page